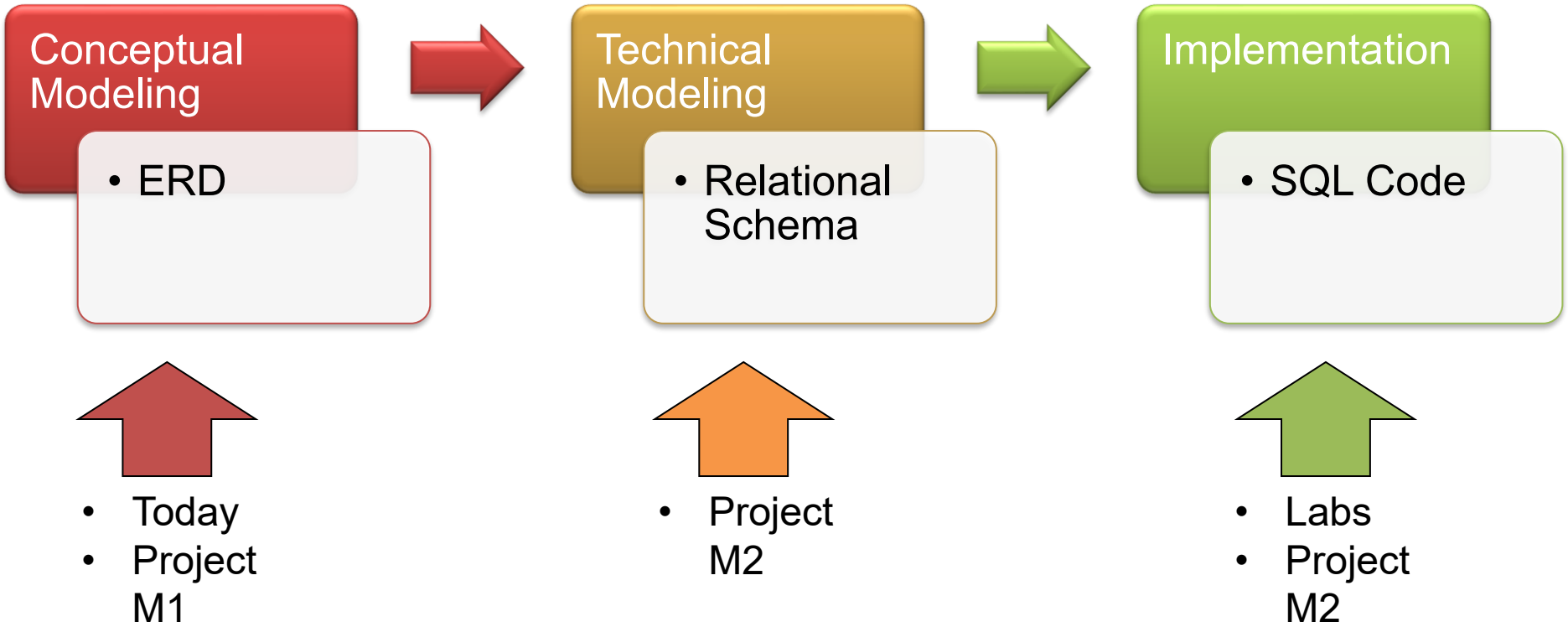
High-Level Database Design

- Entity Relationship Diagram
- ERD Notation
- Example Database Application
- Developing an ERD from textual descriptions and forms

Database Design/Implementation

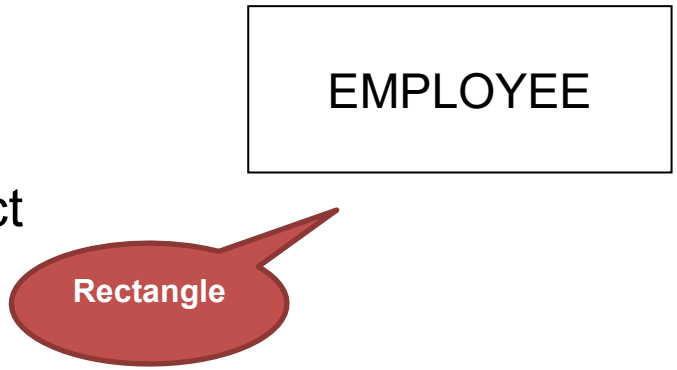


ERD

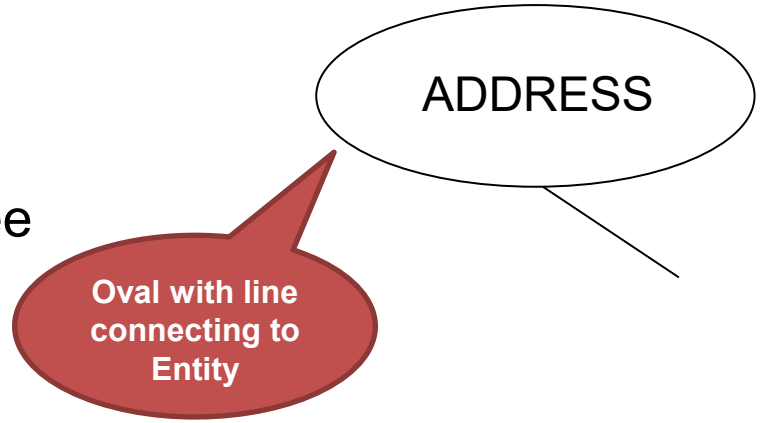
- Entity Relationship Diagram
- Consists of:
  - Entities
  - Attributes
  - Relationships

Entity

- **Entity**: an object (a "thing") of the real world with an independent existence
  - E.g. Employee, Department, Project
  - Usually Name in Singular (not Plural)

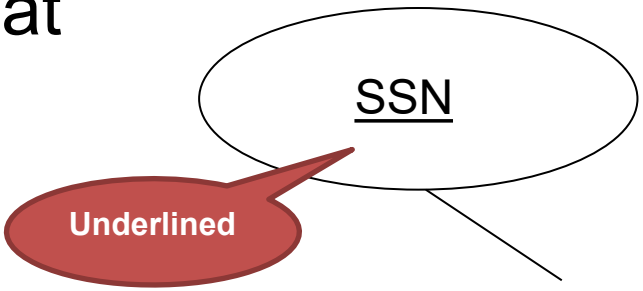


- **Attribute**: A property that describes an entity
  - E.g. Address, Salary of employee

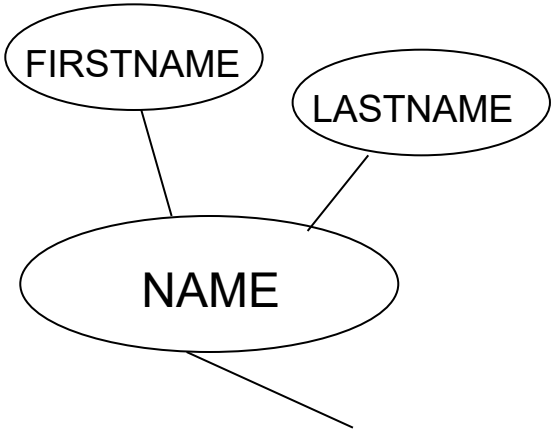


Types of Attributes

- **Key Attribute**: A **property** that uniquely identifies an entity
  - E.g. SSN for Employee

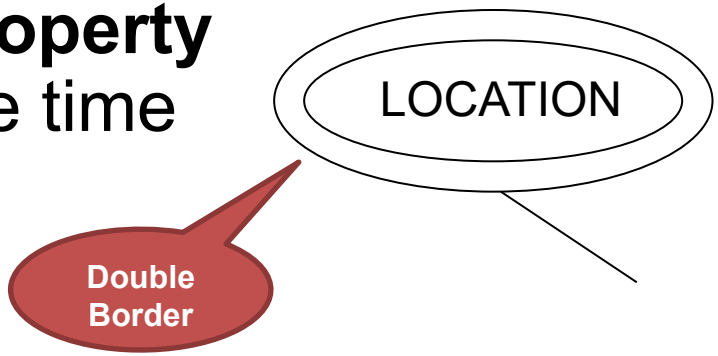


- **Composite Attribute**: A **property** consisting of other properties
  - E.g. Name of Employee

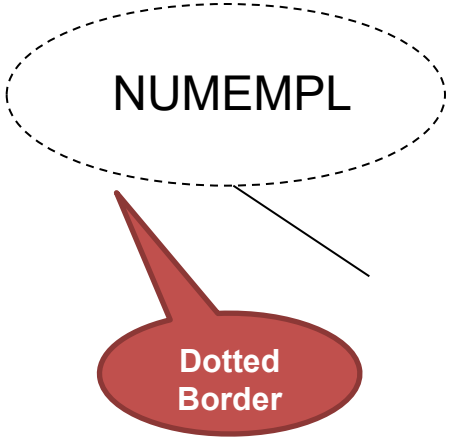


Types of Attributes

- **Multi-valued Attribute**: A **property** with many values at the same time
  - E.g. Locations of DEPARTMENT

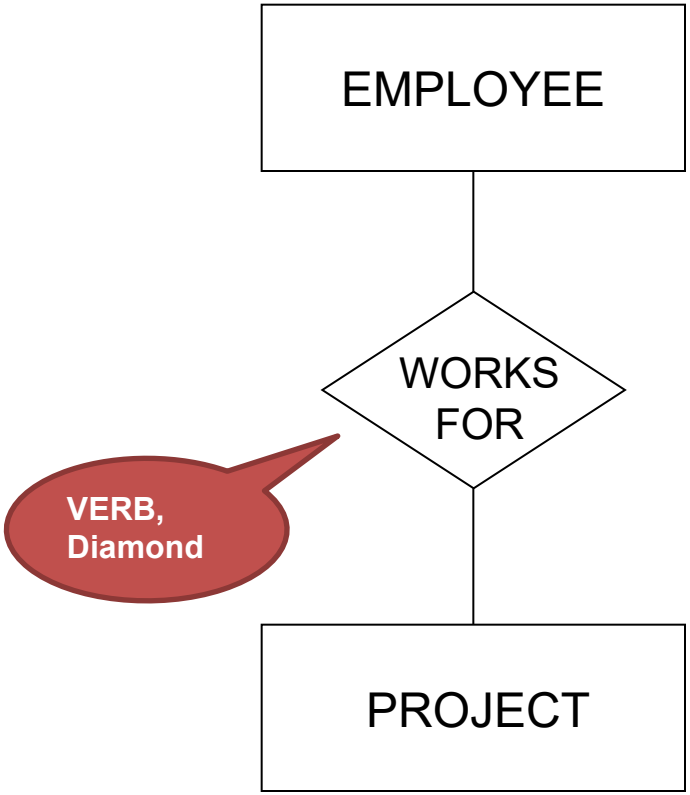


- **Derived Attribute**: A **property** calculated from other properties (not stored)
  - E.g. Number Of Employees of Department



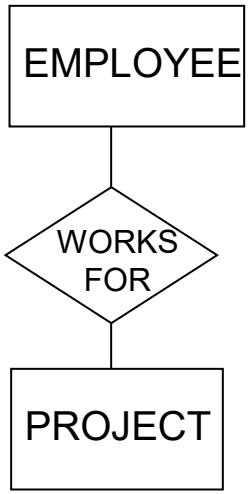
Relationships

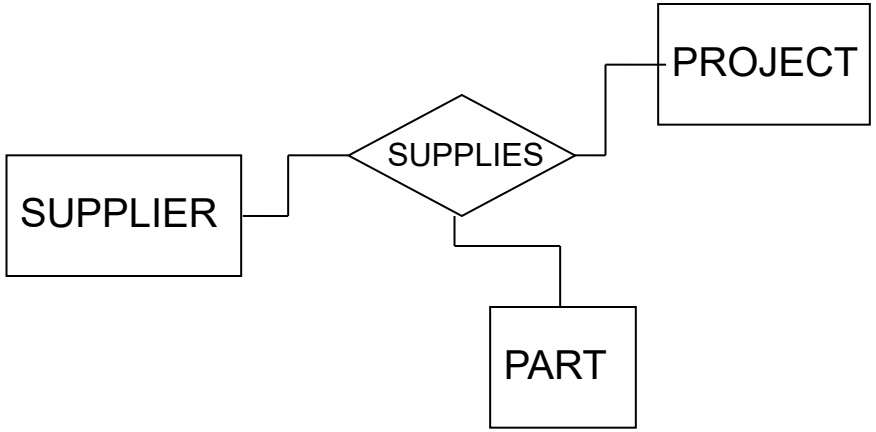
- **Relationship**
  - An association between entities
    - E.g <u>Employee</u> **WORKS FOR** <u>Project</u>
  - ERD notation
    - References represented by a Relationship
      - E.g. Employee WORKS_FOR Project
    - NOT BY ATTRIBUTES!
      - E.g. attribute *Department* of EMPLOYEE refers to a DEPARTMENT entity



Relationship Degree

- number of entities participating in the relationship
  - Employee **WORKS FOR** Project
    - **binary (2)**
  - Supplier **SUPPLIES** a Part to a Project
    - **ternary (3)**





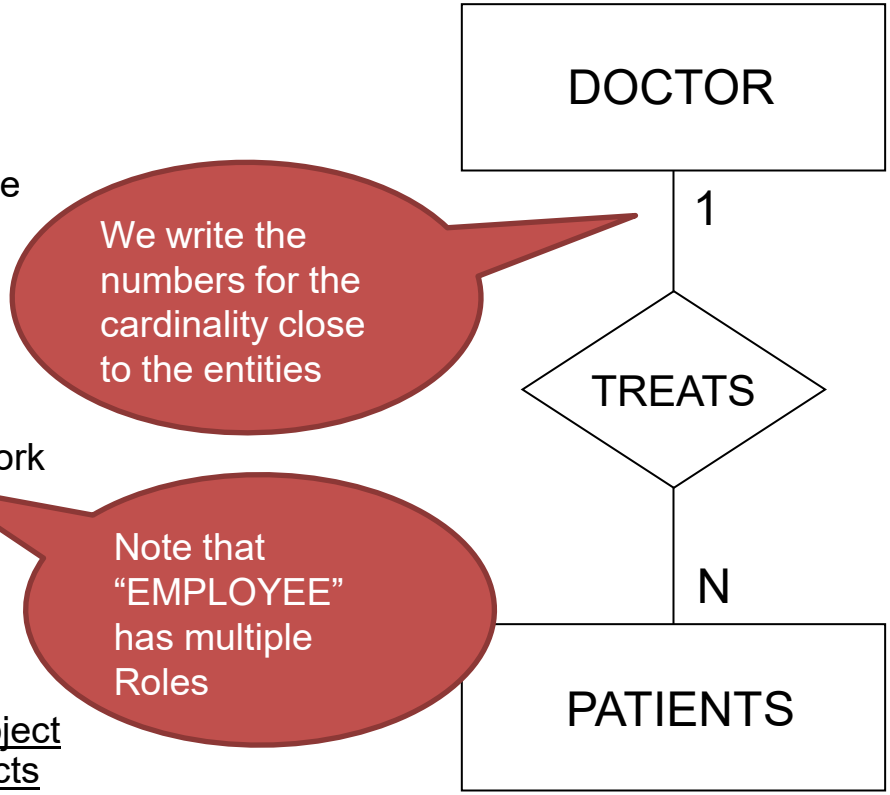
Relationship Roles

- Define meaning of relationship
- **Role Names** explain the role that the entity plays
  - E.g. **WORKS FOR** relationship
    - Employee entity plays the role of the *Worker*
    - Department entity plays the role of the *Employer*
- Not necessary
  - When participating entities are distinct
- Absolutely necessary
  - When same entity participates more than once in the same relationship (recursive relationships)
  - E.g. Employee (*Supervisor*) **SUPERVISES** Employee (*Supervisee*)
  - E.g. Person **IS PARENT OF** Person

Additional label on line close to relevant entity

Cardinality Ratio

- Number of relationship instances an entity can participate in
- One-to-one (1:1)
  - One may be related only to another
  - 1 Department **HAS** 1 Employee with the Role Manager
  - E.g. Supervisor / Team Coach
- One-to-Many (1:N, N:1)
  - One may be related to many
  - 1 Department **HAS** many Employees that work in it
  - E.g. Regular Employee / Team Player
- Many-to-Many (N:M)
  - many may be related to many
  - Many Employees **WORK** on the same 1 Project and 1 Employee can **WORK** on many Projects
  - E.g. Students and Courses

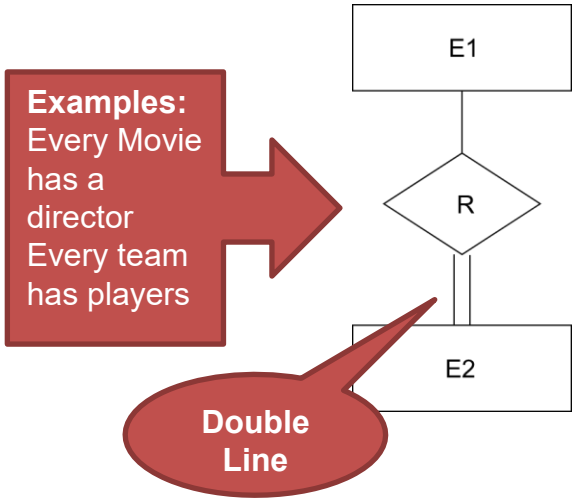


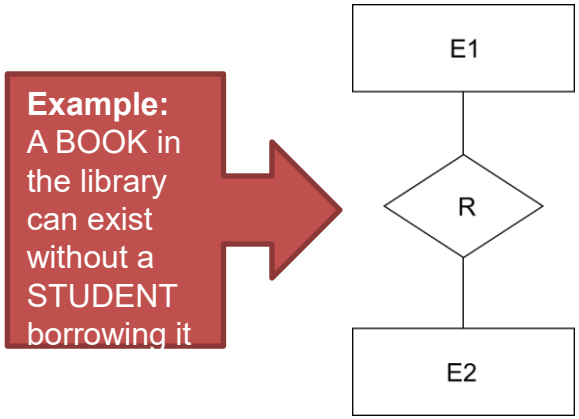
Participation Constraint

- Specifies if existence of entity depends on its relationship to another entity

- **Total participation**
  - "existence dependency" of E2 in R
  - E2 instance cannot exist if not related (through R) to an E1 instance
    - E.g. an Employee cannot exist if he/she does not **WORK FOR** a Department

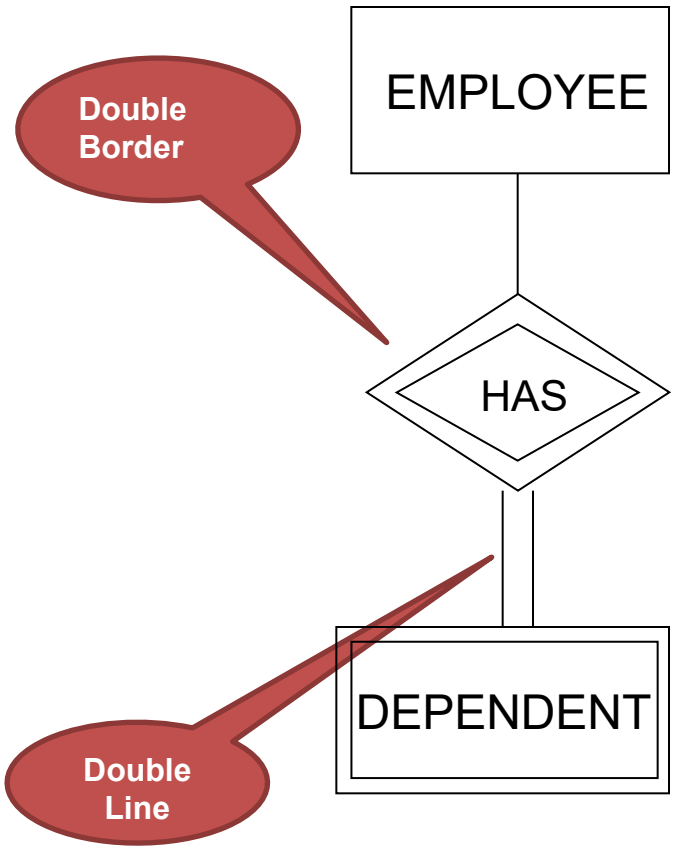
- **Partial participation**
  - E2 instance exists even if not related (through R) to an E1 instance
    - E.g. a Department exists even if it **CONTROLS** no Project





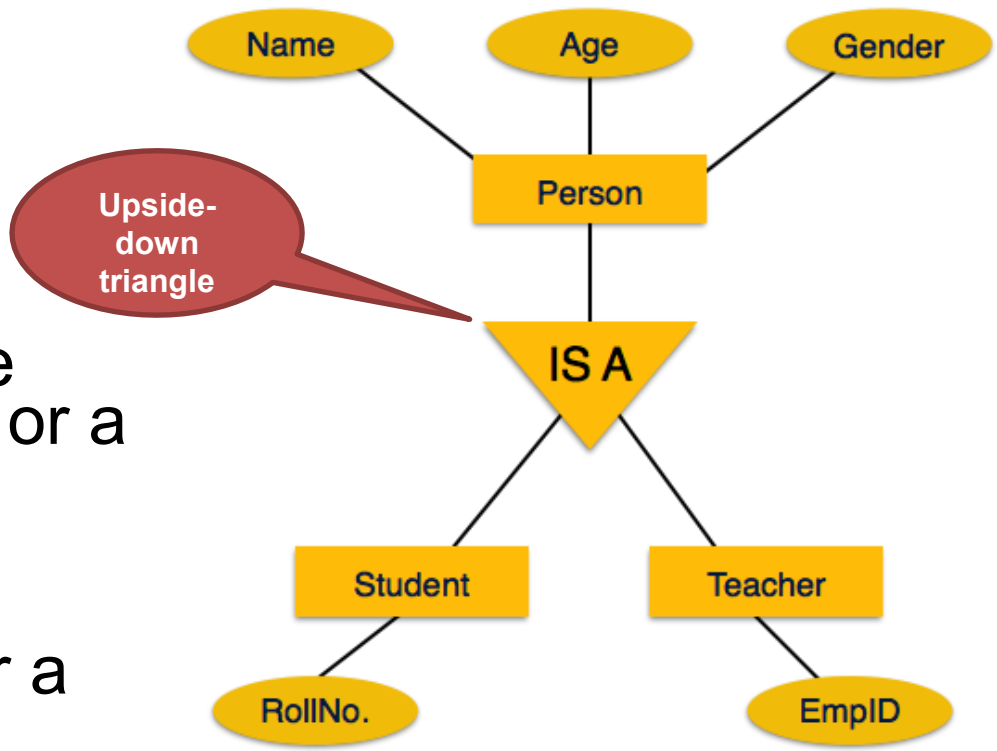
Weak Entities

- Entities with **no Key Attribute** of their own
  - E.g. <u>Dependent</u>
    - Dependent family members have the SSN number of their father/mother
    - The Name of the Dependent family member does not uniquely identify them (“Partial Key”)
    - There may be many George's that are sons of an Employee
- Identified by being related to "**strong**“ entities in combination with some of their attribute values
  - We call the other entity **Identifying** or **Owner**
  - We call the relationship that relates a weak entity to its owner the **Identifying Relationship**
  - The weak entity always has **Total Participation** in the identifying relationship



Hierarchies

- Special kind of relationship
- No attributes
- Depicted with a triangle shape
- Examples
  - A person can be either a student or a teacher
  - A user can be a super user, an administrator, or a regular user



Examples

Example 1 – “The Company”

- “The Company” Database keeps track of the company’s
  - Employees
  - Departments
  - Projects

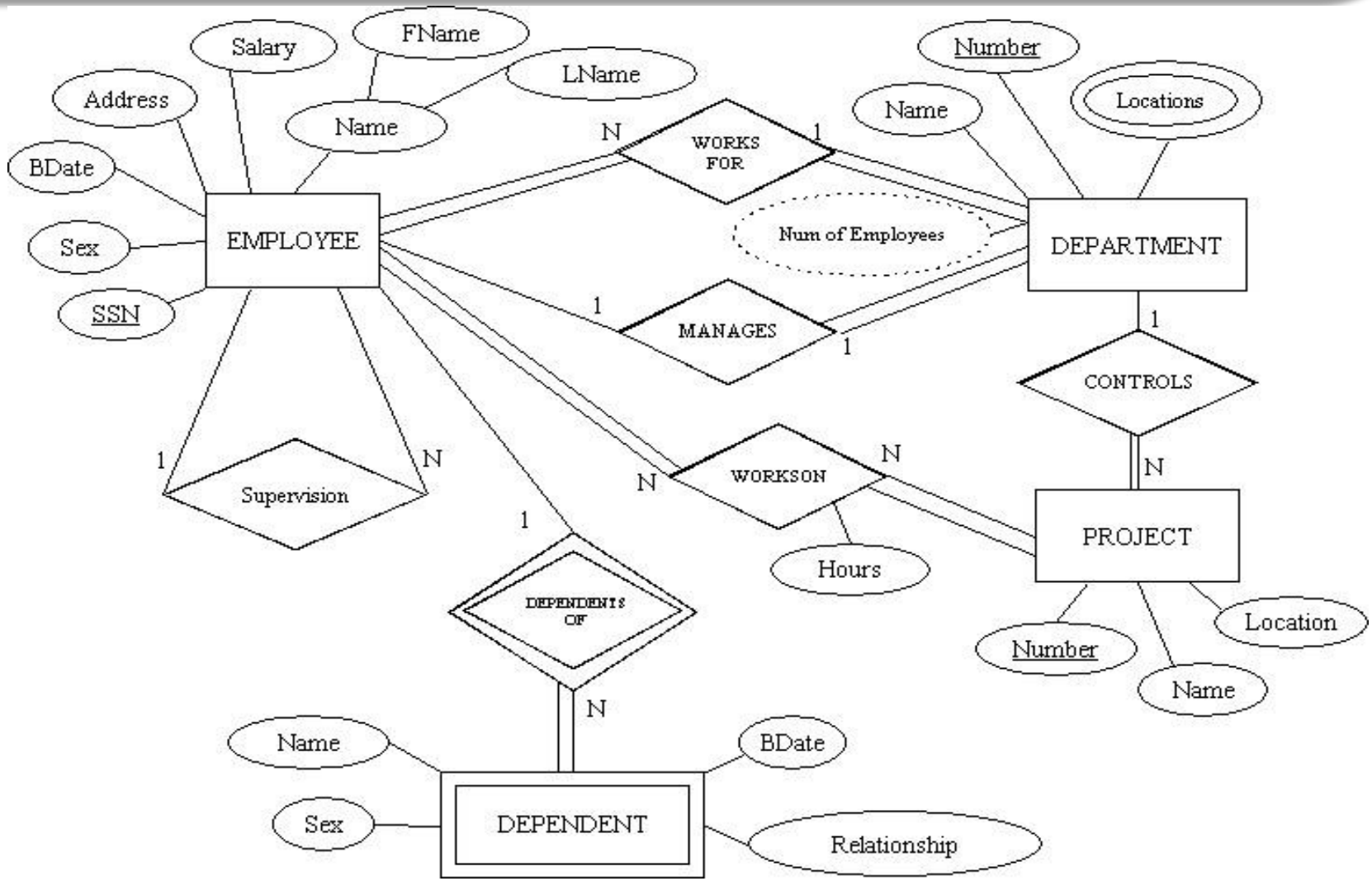
- The company is organized into **Departments**
  - Each department has
    - A unique **Name**
    - A unique **Number**
    - A particular **Employee** who manages the department
    - Several **Locations**
  - We keep track of a **Start Date** when the employee as the manager of the department

- A department controls a number of **Projects**
  - Each project has
    - A unique **Name**
    - A unique **Number**
    - One single **Location**

Example 1 – "The Company"

- An **Employee** has
  - A **Name** (which consists of First and Last) – the name is not necessarily unique
  - A unique **Social Security Number**
  - One **Address**
  - One **Salary**
  - A specific **Sex**
  - A specific **Birthdate**
  - An Employee
    - is assigned to 1 **Department**
    - may work on several **Projects**, which are not necessarily controlled by the same department
  - We keep track of
    - the **Number of Hours** per week that an Employee **WORKS** on each **Project**.
    - the direct **Supervisor** of each Employee
    - the **Dependents** (wife, children) of each employee (for insurance purposes)
- Each **Dependent** has
  - A **First Name**
  - A specific **Sex**
  - A specific **Birthdate**
  - A specific **Relationship** to the Employee (wife, child, etc)

Example 1 - ERD



Example 2 - Stillwater Antiques

"Customer Requirements" *(Design from Description)*

Stillwater Antiques plans to create a database system, named SADB, to keep track of the information about its antique items, clients, and buying and selling transactions. Stillwater buys and sells one-of-a-kind antiques of all kinds (for example, furniture, jewelry, china, and clothing). Each item is assigned a unique item number and is also characterized by a description, asking price, and condition.

Stillwater works with many different individuals, called clients, who sell items to and buy items from the store. Some clients only sell items to Stillwater, some only buy items, and some others both sell and buy. Each client is assigned a unique client number and is also described by a client name and client address.

When Stillwater sells an item in stock to a client, it records the actual selling price (may be different from the asking price), sales tax (tax of zero indicates a tax exempt sale), and date sold. When Stillwater buys an item from a client, it records the purchase cost, date purchased, and condition at time of purchase.

Example 2 - Stillwater Antiques

- DB keeps track of information about its
  - antique **Items**
  - **Clients**
  - **Buying** and **Selling Transactions**

- **Items** are one-of-a-kind antiques of all kinds (furniture, jewelry, china, clothing), each having
  - A unique **Item Number**
  - A **Description**
  - An **Asking Price**
  - A specific **Condition**
- **Clients** sell and buy items from the store
  - Some only sell items, some only buy items, and some others both sell and buy
  - Each client has
    - A unique **Client Number**
    - A **Description**
    - An **Address**

- **Selling Transaction** records
  - The actual **Selling Price** (may be different from the asking price)
  - The **Sales Tax** (tax of zero indicates a tax exempt sale)
  - The **Date Sold**
- **Buying Transaction** Records
  - the **Purchase Cost**
  - the **Date Purchased**
  - the **Condition** at time of purchase

<u>Example</u>

Stillwater bought item 5020, a round oak table, from client 2012, Mr. Brown, at $300 on February 20, 1999, and old it to client 1092, Ms. Lin, for $750 on March 23, 2000. The asking price was $900

Example 3 - Covid Hospital

"Customer Requirements" *(Design from Examples)*

"Orange Form" - a statement given to the customer (patient)

Covid Hospital
Account Statement

Patient name: Hamilton, Lewis
Patient address: 300 London Str.
City-State-Zip: Mountain View, Co. 80638

Patient#: 3249
Date admitted: 1/12/2020
Date discharged: 3/12/2020

Item Code	Description	Charge
200	Rom semi-pr	150
205	Television	10
307	X-ray	25
413	Lab tests	35
	Balance Due	220

Example 3 - Covid Hospital

"Blue Form" - a doctor reporting all procedures performed in a day

Covid Hospital
Physician Report

Date: 3/1/2020

Physician ID: Doktorakos
Physician Phone: 329-1848

Patient#	Patient Name	Location	Procedure
6083	Mitsos, Kitsaras	184-2	Stupidoectomy
3157	Marika, Nosobright	216-1	Smart Brain Transplant
4139	Eyeam, Doomed	107-3	Vitaminization to endure Weekend classes